

K.1 The video compositor

```

+-----+
| HDMI / display output |
+-----+
      ^
      |
+-----+
| Video compositor      |
| (per-pixel layer mux, |
| palette, blend modes) |
+-----+
+--+--+--+--+--+--+--+
|  |  |  |  |  |  |  |
layer 20 |  |  |  |  |  |  | (front)
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  | +-- Voodoo 3D
|  |  |  |  |  |  |  | +----- ULA
|  |  |  |  |  |  |  | +----- ANTIC + GTIA
|  |  |  |  |  |  |  | +----- TED video
|  |  |  |  |  |  |  | +----- VGA
+-----+ VideoChip (layer 0, back)

```

K.2 The audio mixer

The audio mixer has separate engine inputs, plus the SFX channels, and one stereo output stream.

```

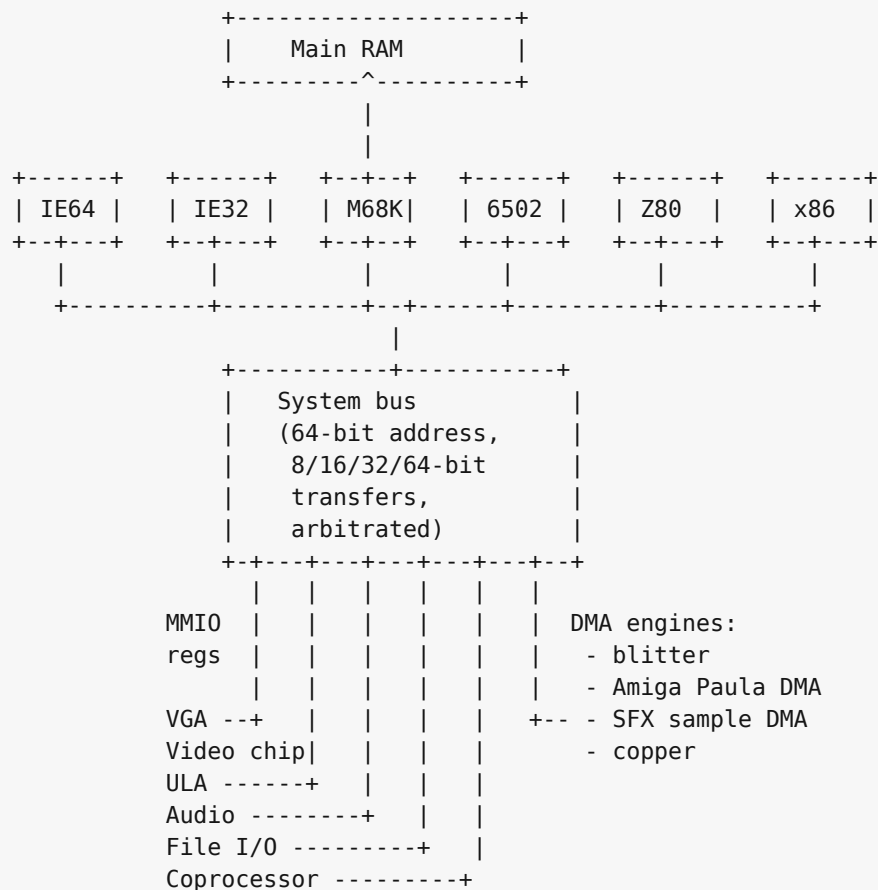
SoundChip --+
PSG / AY --+
SN76489 ---+
SID -----+
SID2 -----+
SID3 -----+
POKEY -----+---> Mixer ---> Overdrive ---> Filter ---> Reverb ---> Output
TED audio -+      (sum,      (global)      (global)      (global)
AHX -----+      per-chip    voice)
MIDI/MUS --+      gain,
MOD -----+      per-chip
WAV -----+      mute)
Paula DMA -+
          |
SFX ch 0-3 +

```

The SoundChip's own filter, the SID family's resonant filter, and the engine-internal effects all feed the mix before the global overdrive / filter / reverb stage; the global effects apply once per output sample to the summed signal.

K.3 The system bus

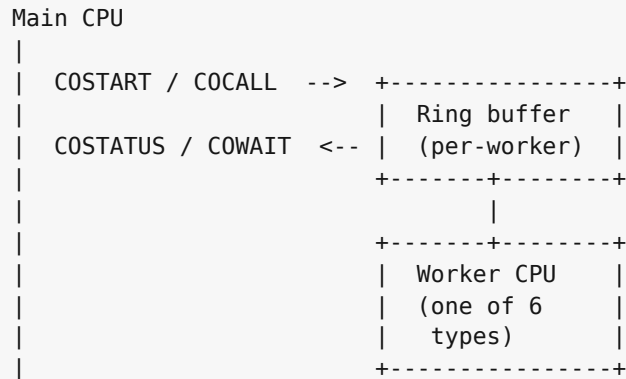
Every CPU and every device hangs off one shared 64-bit physical bus. The bus carries 64-bit physical addresses and supports 8, 16, 32, and 64-bit transfers through CPU and device adapters. It arbitrates simultaneous accesses on a round-robin basis; there is no priority encoding above the CPU / DMA distinction.



IE32, M68K, and x86 are 32-bit bus clients. The 8-bit CPUs (6502, Z80) reach the bus through an address translator that turns their 16-bit address space into selected low-window bus addresses, with the bank registers described in Chapters 27 and 28 selecting which translation applies.

K.4 Coprocessor channels

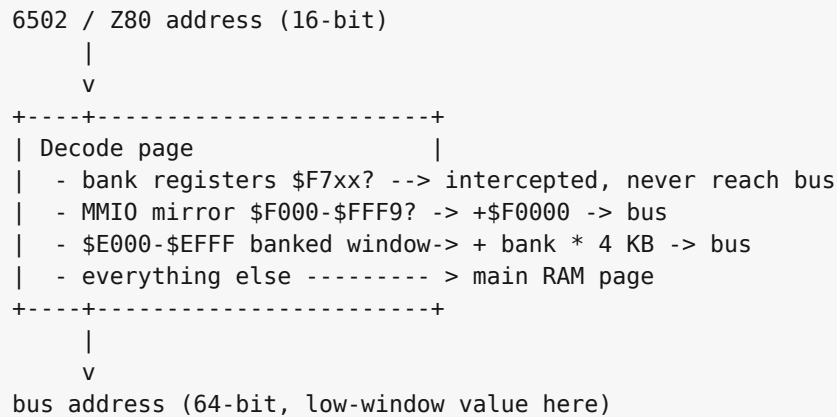
The coprocessor block (Chapter 32) is a many-to-many channel between the main CPU and a pool of worker CPUs. Each worker listens on its own ring buffer; the main CPU posts work items into the ring and reads the result back.



Six worker types share the protocol: an IE64 worker, an IE32 worker, a 6502 worker, a Z80 worker, an M68K worker, and an x86 worker. The main CPU is whichever one boots the machine; all the others are available as workers when not booted.

K.5 Bus translation for the small CPUs

The 6502 and Z80 see the same overall bus through a 16-bit-to-64-bit translator. In practice their documented apertures target the low memory and MMIO window. The translator has two independent functions:



The bank registers at \$F700-\$F705 (low/hi pairs for apertures 1, 2, 3) and \$F7F0 (aperture 0) are captured before the translator runs. A write to \$F700 does not reach \$F0700; it loads the low byte of bank register 1.